

Proximal Gradient Algorithm

$$\min: f(x) = \underset{\substack{\uparrow \\ \text{smoother}}}{g(x)} + \underset{\substack{\uparrow \\ \text{easy to evaluate}}}{h(x)}$$

prox

$$\underline{x_{t+1} = \text{prox}_{\eta h}(x_t - \eta \nabla g(x_t))}$$

// if $\boxed{h(x) = \mathcal{I}_{\mathcal{X}}(x)}$ then
prox-gradient becomes
projected gradient.

Properties of Prox

Recall that for projection: $\|y_1 - y_2\|_2 \geq \|\text{Proj}_X(y_1) - \text{Proj}_X(y_2)\|_2$

Similarly: $\|\text{prox}_h(x) - \text{prox}_h(y)\|_2 \leq \|x - y\|_2$. $\leftarrow$ follows from $\hookrightarrow$ using C-S.

$$\langle \text{prox}_h(x) - \text{prox}_h(y), x - y \rangle \geq \|\text{prox}_h(x) - \text{prox}_h(y)\|_2^2$$

let $u = \text{prox}_h(x)$ $v = \text{prox}_h(y)$

$$u = \text{prox}_h(x) = \underset{u}{\text{argmin}}: h(u) + \frac{1}{2} \|x - u\|_2^2$$
$$0 \in \partial h(u) + (u - x)$$

$$\Leftrightarrow \boxed{(x - u) \in \partial h(u)} \quad \text{also} \quad \boxed{(y - v) \in \partial h(v)}$$

h convex $\Rightarrow \partial h$ monotone

$$\Rightarrow \langle (x - u) - (y - v), u - v \rangle \geq 0$$

$$\Rightarrow \langle x - y, u - v \rangle \geq \|u - v\|_2^2$$

$$\Rightarrow \langle x - y, \text{prox}_h(x) - \text{prox}_h(y) \rangle \geq \|\text{prox}_h(x) - \text{prox}_h(y)\|_2^2.$$

Proximal Gradient - Gradient mapping

For smooth functions, key inequality:

For GD of smooth f g - $x_{t+1} = x_t - \eta \nabla g(x_t)$

$$g(x_{t+1}) = g(x_t - \frac{1}{\beta} \nabla g(x_t)) \leq g(x_t) - \frac{1}{2\beta} \|\nabla g(x_t)\|_2^2 \quad (\text{from quadratic upper bound})$$

$$\text{Now: } x_{t+1} = \text{prox}_{\eta h}(x_t - \eta \nabla g(x_t))$$

$$\min: f(x) = g(x) + h(x)$$

The gradient mapping: Def'n: $G_\eta(x) = \frac{1}{\eta} (x - \text{prox}_{\eta h}(x - \eta \nabla g(x)))$

This gives: $x_{t+1} = x_t - \eta G_\eta(x_t)$

Note: $G_\eta(x_t)$ in general is not $\in \partial f(x_t)$

Proximal Gradient

Recall:

$$G_\eta(x) = \frac{1}{\eta} (x - \text{prox}_{\eta h}(x - \eta \nabla g(x)))$$

Optimal solutions are the only fixed points of the prox grad update

$$x_{t+1} = x_t - \eta G_\eta(x_t)$$

equivalently: $G_\eta(x) = 0 \iff x$ minimizes $f(x) = g(x) + h(x)$

To see this:

$$G_\eta(\hat{x}) = 0 \iff \hat{x} = \text{prox}_{\eta h}(\hat{x} - \eta \nabla g(\hat{x}))$$

$$\iff (\hat{x} - \eta \nabla g(\hat{x})) - \hat{x} \in \eta \partial h(\hat{x})$$

$u = \text{prox}_{\eta h}(x)$: $0 \in \partial h(u) + \frac{1}{\eta}(u - x)$
 $(x - u) \in \eta \partial h(u)$

$$\iff -\eta \nabla g(\hat{x}) \in \eta \partial h(\hat{x})$$

$$\iff \boxed{0 \in \nabla g(\hat{x}) + \partial h(\hat{x})}$$

$$\min: g(x) + h(x) \quad \hat{x} \text{ opt sol'n} \iff 0 \in \nabla g(\hat{x}) + \partial h(\hat{x})$$

Proximal Gradient - key lemma

Need analog: $g(x - \frac{1}{\beta} \nabla g(x)) \leq g(x) - \frac{1}{2\beta} \|\nabla g(x)\|_2^2$.

Key lemma:

$f = g + h$, g is β -smooth, α -strongly convex, (α may be zero)

$$\eta \leq \frac{1}{\beta}$$

Lemma: $f(x - \eta G(x)) \leq f(z) + \langle G_\eta(x), x - z \rangle - \frac{1}{2} \|G_\eta(x)\|_2^2 - \frac{\alpha}{2} \|x - z\|_2^2$.

Consequence: analog of our basic inequality

Use Lemma with: $z = x_t$, $x = x_t$

$$f(x_{t+1}) = f(x_t - \eta G(x_t)) \leq f(x_t) - \frac{1}{2} \|G_\eta(x_t)\|_2^2$$

compare to inequality at top.